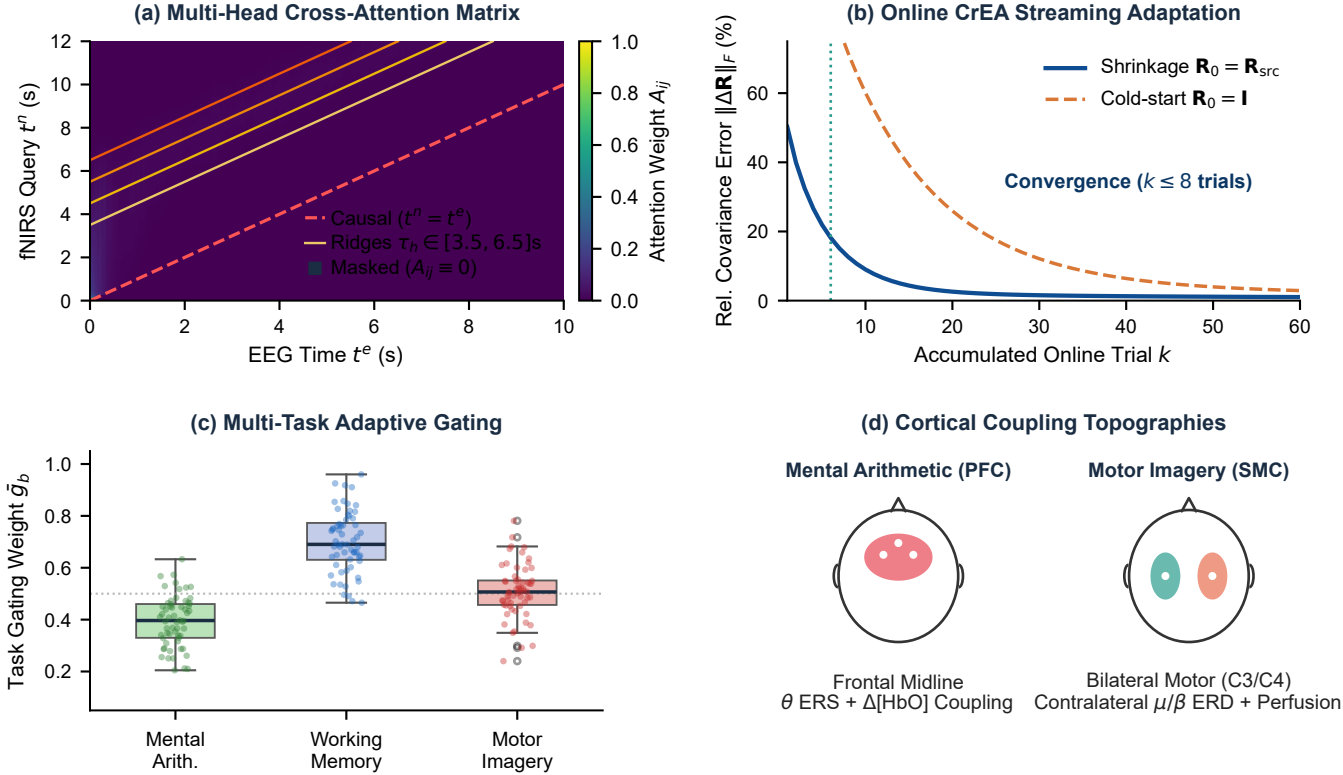

(a) Multi-Head Cross-Attention Matrix
fNIRS Query t^n (s)
EEG Time t^e (s)
Attention Weight A_{ij}
Causal ($t^n = t^e$)
Ridges $\tau_h \in [3.5, 6.5]$s
Masked ($A_{ij} \equiv 0$)
(b) Online CrEA Streaming Adaptation
Rel. Covariance Error $\|\Delta\mathbf{R}\|_F$ (%)
Accumulated Online Trial k
Shrinkage $\mathbf{R}_0 = \mathbf{R}_{\text{src}}$
Cold-start $\mathbf{R}_0 = \mathbf{I}$
Convergence ($k \leq 8$ trials)
(c) Multi-Task Adaptive Gating
Task Gating Weight $\bar{g}_b$
Mental Arith.
Working Memory
Motor Imagery
(d) Cortical Coupling Topographies
Mental Arithmetic (PFC)
Motor Imagery (SMC)
Frontal Midline
θ ERS + Δ[HbO] Coupling
Bilateral Motor (C3/C4)
Contralateral μ/β ERD + Perfusion